

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 311 — Past Questions & Materials
by Renance

Course: GET 311
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 8
Levels: 300

This study pack was compiled and restructured by Renance for GET 311 students at Federal University of Petroleum Resources, Effurun. It brings together 8 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 8 — GET 311 Engineering Mathematics III

Level 300 • First Semester • PastQuestion

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| “ ee COLLEGE OF ENGINEERING AND TECHNOLOGY

Bib Eat, IRST SEMESTER EXAMINATION - 2024/2025 ACADEMIC SESSION

COURSE CODE AND TITLE: GET 311 - ENGINEERING MATHEMATICS III (3 UNITS)

INSTRUCTION: ANSWER FOUR (4) QUESTIONS WITH AT LEAST ONE (1) FROM EACH SECTION

TIME ALLOTTED: 3 HOURS

SECTION A

QUESTION 1 [17.5 MARKS]

Find the Fourier series of the function, $f(x) = x$?

QUESTION 2 [17.5 MARKS]

Find the Taylor series for

a) $f(x) = \sqrt{1+x}$ centered around zero

b) $f(x) = \sqrt{1+x}$ centered around zero

ON- $T < x < T_r$

SECTION B

QUESTION 3 [17.5 MARKS] Using the Laplace transform method

a). Solve the equation $\frac{d^2y}{dt^2} - 2y = 4$ given that at $t=0$, $y=1$.

resistor of

b). an inductor of 2 henrys, a $\frac{1}{2}$ 16 ohms and a capacitor of 0.2 farads are connected in series with an e.m.f. of E volts. At $t=0$ the charge on the capacitor and current in the circuit are zero. Find the charge and current at any time $t > 0$ if $E = 300$

2h 5 _t

oie : __T 0.8 fd

16 ohms

QUESTION 4 [17.5 MARKS]

a). Define the z-transform of a Sequence $x(n)$, and find its transform given that $x(n) = a^n u(n)$.

b). A pipeline system experiences a sudden

governing equation for pressure

b are constants, and the in

pressure drop due to a Valve closure. The

. ' " : dp _

$p(t)$ in the pipeline is given by $ae + @P = b$ where a and

itial pressure before closure is 1 unit. Determine $p(t)$.

“ee

SECTION C

QUESTION 5 [17.5 MARKS]

An optimization project in a data center had necessitated the modelling of cooling rate of battery banks, resulting in the following Legendre equation, which relates the surface temperature (T , in degrees Celsius) of a battery to its duration of exposure (t , in seconds) to forced cooling media. Obtain the recurrence relation and find two linearly independent solutions of the modelling Legendre equation. Consequently, determine any independent solutions that could be assumed after 5 seconds of its exposure to the cooling media.

1- ϕ ? at $9T = 0$

1-0) Give at -

QUESTION 6 [17.5 MARKS]

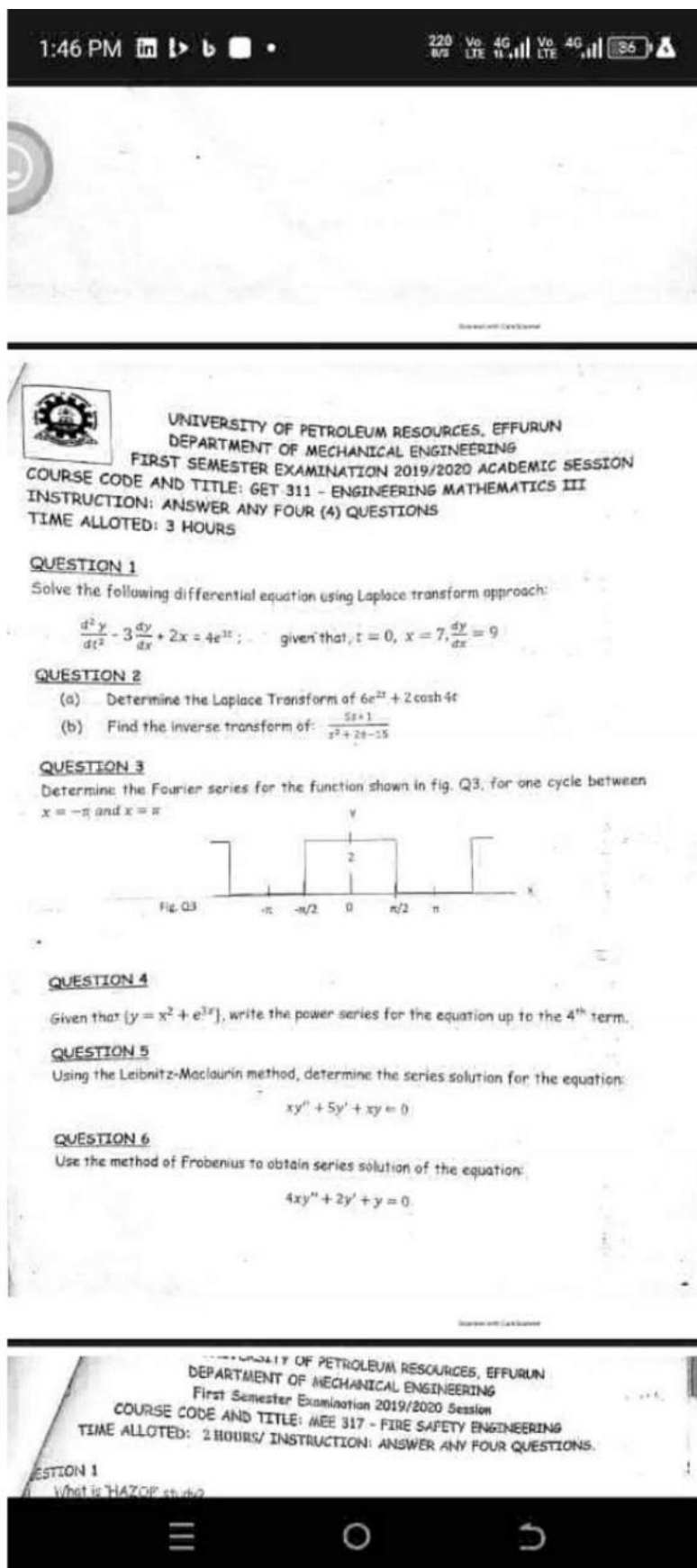
For the modelling equation relating the depth of an oil well (x) with crude quantity in the well (y) shown below, verify the applicability of the Frobenius method to solve the model. If applicable, find two linearly independent solutions about the origin.

the modelling equation

Paper 2 of 8 — GET 311 General Mathematics 111

Level 300 • First Semester • PastQuestion

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Paper 3 of 8 — GET 311 ENGINEERING MATHEMATICS III 2024/2025

Level 300 • First Semester • PastQuestion

SECTION A

QUESTION 1 [17.5 MARKS]

Find the Fourier series of the function, $F(x) = x$ on $-\pi < x < \pi$

Find the Fourier series of the function, $F(x) = x$ on $-\pi < x < \pi$

QUESTION 2 [17.5 MARKS]

Find the Taylor series for

a) $f(x) = \sin x$ centered around zero

b) $f(x) = \cos x$ centered around zero

Capacitor of .02 farads are

On the charge on the Capacitor and

current at any time $t > 0$ if $E = 300$

16 ohms

Y

QUESTION 4 [17.5 MARKS]

a). Define the z-transform of a sequence $x(n)$, and find its transform given that $x(n) = a^n u(n)$.

b). A pipeline system experiences a sudden pressure drop. The governing equation for pressure $p(t)$ in the pipeline is given by $m \frac{dp}{dt} + ap = b$ where a and b are constants, and the initial pressure before closure is 1 unit. Determine $p(t)$.

b). A pipeline system experiences a sudden pressure drop.

The governing equation for pressure $p(t)$ in the

pipeline is given by $m \frac{dp}{dt} + ap = b$ where a and b

are constants, and the initial pressure before closure is 1 unit. Determine $p(t)$.

initial pressure before closure is 1 unit. Determine $p(t)$.

SECTION B

QUESTION 5 [17.5 MARKS]

An optimization project in a data center had necessitated the modelling of cooling rate of battery banks, resulting in the following Legendre equation, which relates the surface temperature (T , in degrees Celsius) of a battery to its duration of exposure (t , in seconds) to forced cooling media. Obtain the recurrence relation and find two linearly independent solutions of the modelling Legendre equation. Consequently, determine any two approximate values the battery's surface temperature could assume after 95 seconds of its exposure to the cooling media.

$(1 - t^2) \frac{d^2 T}{dt^2} + 2t \frac{dT}{dt} = 0$

$(1 - t^2) \frac{d^2 T}{dt^2} + 2t \frac{dT}{dt} = 0$

$(1 - t^2) \frac{d^2 T}{dt^2} + 2t \frac{dT}{dt} = 0$

QUESTION 6 [17.5 MARKS] . |

For the modelling equation relating the depth of an oil well (x) with crude quantity in the well (y) shown below, verify the applicability of the Frobenius method to solve the model. If applicable, find two linearly independent solutions of the modelling equation about the origin.

$$xy'' + 2y' + y = 0$$

Paper 4 of 8 — GET 311 Engineering Mathematics 2024/2025

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY

[RST SEMESTER EXAMINATION - 2024/2025 ACADEMIC SESSION

SECTION

TIME ALLOTTED: 3 HOURS

SECTION A

QUESTION 1 (17.5 MARKS]

Find the Fourier series of the function, $F(x) = x^2$

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QUESTION 2 [17.5 marks] t wa Pt itl :

Find the Taylor series for ¥ biwoe val en yng n

a) $F(x) = \sqrt{1+4x}$ centered around zero 4) = a

ipre) L? Y-2. (pont l

b) $f(x) = \sqrt{1+x}$ centered around zero yolrece (he) zy z | (

SECTICN B aE G bed =(408-= 1 AKANE oF

QUESTION 3 [17.5 MARKS] Usin

g the LaPlace transform method ce ies

a). Solve the equation $y'' - 2y' =$

4 given that at $x=0$, $y=1$, a & $(Fy) \times a$

b). an inductor of 2 henrys, a resistor of 16 ohms and a capacitor of .02 farads are connected in series with an emf, of E volts. At $t=0$ the charge on the capacitor and current in the circuit are zero. Find the charge and current at any time $t > 0$ if

4

= 300

(volts) here lK le

: Ur) See

2h == 02 fd

L AANA

16 otras

UESTION 4 [17.5 MARKS]

a). Define the z-transform of ase

a'u(n).

b). A pipeline system experiences a sudden pressure drop ae a Valve closure. The

governing equation for pressure $p(t)$ in the pipeline is given by $- \dot{p} + ap = b$ where a and b are constants, and the initial pressure before closure is 1 unit. Determine $p(t)$.

quence $x(n)$, and find its transform given that $x(n) =$

n Y

arers

SECTION C

QUESTION 5 [17.5 MARKS]

An optimization project in a data center had necessitated the modelling of cooling rate of battery banks, resulting in the following Legendre equation, which relates the surface temperature (T , in degrees Celsius) of a battery to its duration of exposure (t , in seconds) to forced cooling media. Obtain the recurrence relation and find two linearly independent solutions of the modelling Legendre equation. Consequently, determine any two approximate values the battery's surface temperature could as seconds of its exposure to the cooling media.

aT

$(1 - t^2) \frac{d^2 T}{dt^2} + 2t \frac{dT}{dt} = 0$

ae dt i

sume after 5

QUESTION 6 [17.5 MARKS]

For the modelling equation relating the depth of an oil well (x) with crude quantity in the well (y) shown below, verify the applicability of the Frobenius method to solve the model. If applicable, find two linearly independent solutions of the modelling equation about the origin.

$$4xy'' + 2y' + y = 0$$

Paper 5 of 8 — GET 311 ENGINEERING MATHEMATICS III PQ

Level 300 • First Semester • PastQuestion

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Semester (6-67/8578/200)

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY

FIRST SEMESTER EXAMINATION - 2024/2025 ACADEMIC SESSION

COURSE CODE AND TITLE: GET 311 - ENGINEERING MATHEMATICS III (3 UNITS)

INSTRUCTION: ANSWER FOUR (4) QUESTIONS WITH AT LEAST ONE (1) FROM EACH SECTION

TIME ALLOTTED: 3 HOURS

SECTION A

QUESTION 1 [17.5 MARKS]

Find the Fourier series of the function, $f(x) = x^2$ on $-\pi < x < \pi$

QUESTION 2 [17.5 MARKS]

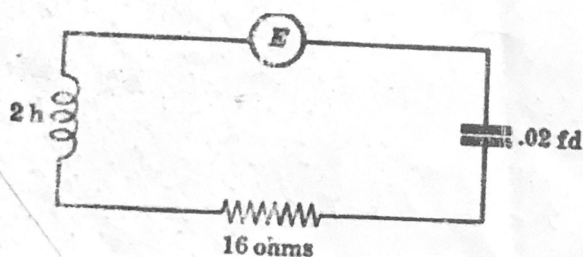
Find the Taylor series for

- $f(x) = \sqrt{1+x}$ centered around zero
- $f(x) = \sqrt[3]{1+x}$ centered around zero

SECTION B

QUESTION 3 [17.5 MARKS] Using the LaPlace transform method

- Solve the equation $\frac{dy}{dt} - 2y = 4$ given that at $t=0$, $y=1$.
- an inductor of 2 henrys, a resistor of 16 ohms and a capacitor of .02 farads are connected in series with an e.m.f. of E volts. At $t_0 = 0$ the charge on the capacitor and current in the circuit are zero. Find the charge and current at any time $t > 0$ if $E = 300$ (volts).



$$\frac{dL}{dt} + 0.02 = 300$$

$$5T^2$$

QUESTION 4 [17.5 MARKS]

- Define the z-transform of a sequence $x(n)$, and find its transform given that $x(n) = a^n u(n)$.
- A pipeline system experiences a sudden pressure drop due to a Valve closure. The governing equation for pressure $p(t)$ in the pipeline is given by $\frac{dp}{dt} + ap = b$ where a and b are constants, and the initial pressure before closure is 1 unit. Determine $p(t)$.

$$sL[s(t)] = s(0)$$

$$sf(t) = \dots$$

$$y' = \dots$$

$$(0+3) \cdot 1 \cdot 29 \cdot t^0$$

SECTION CQUESTION 5 [17.5 MARKS]

An optimization project in a data center had necessitated the modelling of cooling rate of battery banks, resulting in the following Legendre equation, which relates the surface temperature (T , in degrees Celsius) of a battery to its duration of exposure (t , in seconds) to forced cooling media. Obtain the recurrence relation and find two linearly independent solutions of the modelling Legendre equation. Consequently, determine any two approximate values the battery's surface temperature could assume after 5 seconds of its exposure to the cooling media.

$$(1 - t^2) \frac{d^2 T}{dt^2} - 3t \frac{dT}{dt} + 9T = 0$$

QUESTION 6 [17.5 MARKS]

For the modelling equation relating the depth of an oil well (x) with crude quantity in the well (y) shown below, verify the applicability of the Frobenius method to solve the model. If applicable, find two linearly independent solutions of the modelling equation about the origin.

$$4xy'' + 2y' + y = 0$$

$$3(1) a_1 t^1$$

$$3 a_1 t^1$$

$$y'' = 5 L(y) - y(0)$$

$$\sum_{n=0}^{\infty} a_n t^n$$

Paper 6 of 8 — GET 311 engineering mathematics III 2025

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY

FIRST SEMESTER EXAMINATION - 2024/2025 ACADEMIC SESSION

COURSE CODE AND TITLE: GET 311 - ENGINEERING MATHEMATICS III (3 UNITS)

INSTRUCTION: ANSWER FOUR (4) QUESTIONS WITH AT LEAST ONE (1) FROM EACH SECTION

TIME ALLOTTED: 3 HOURS

SECTION A

QUESTION 1 [17.5 MARKS]

Find the Fourier series of the function, $f(x) = x^2$ on $-\pi < x < \pi$

QUESTION 2 [17.5 MARKS]

Find the Taylor series for

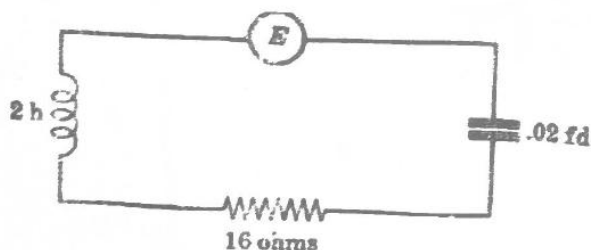
a) $f(x) = \sqrt{1+x}$ centered around zero

b) $f(x) = \sqrt[3]{1+x}$ centered around zero

SECTION B

QUESTION 3 [17.5 MARKS] Using the LaPlace transform method

- a). Solve the equation $\frac{dy}{dt} - 2y = 4$ given that at $t=0$, $y=1$.
- b). an inductor of 2 henrys, a resistor of 16 ohms and a capacitor of .02 farads are connected in series with an e.m.f. of E volts. At $t_0 = 0$ the charge on the capacitor and current in the circuit are zero. Find the charge and current at any time $t > 0$ if $E = 300$ (volts).



QUESTION 4 [17.5 MARKS]

- a). Define the z-transform of a sequence $x(n)$, and find its transform given that $x(n) = a^n u(n)$.
- b). A pipeline system experiences a sudden pressure drop due to a Valve closure. The governing equation for pressure $p(t)$ in the pipeline is given by $\frac{dp}{dt} + ap = b$ where a and b are constants, and the initial pressure before closure is 1 unit. Determine $p(t)$.

SECTION CQUESTION 5 [17.5 MARKS]

An optimization project in a data center had necessitated the modelling of cooling rate of battery banks, resulting in the following Legendre equation, which relates the surface temperature (T , in degrees Celsius) of a battery to its duration of exposure (t , in seconds) to forced cooling media. Obtain the recurrence relation and find two linearly independent solutions of the modelling Legendre equation. Consequently, determine any two approximate values the battery's surface temperature could assume after 5 seconds of its exposure to the cooling media.

$$(1 - t^2) \frac{d^2 T}{dt^2} - 3t \frac{dT}{dt} + 9T = 0$$

QUESTION 6 [17.5 MARKS]

For the modelling equation relating the depth of an oil well (x) with crude quantity in the well (y) shown below, verify the applicability of the Frobenius method to solve the model. If applicable, find two linearly independent solutions of the modelling equation about the origin.

$$4xy'' + 2y' + y = 0$$

Paper 7 of 8 — GET 311 ENGINEERING MATHEMATICS III 2025

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY

FIRST SEMESTER EXAMINATION - 2024/2025 ACADEMIC SESSION

COURSE CODE AND TITLE: GET 311 - ENGINEERING MATHEMATICS III (3 UNITS)
INSTRUCTION: ANSWER FOUR (4) QUESTIONS WITH AT LEAST ONE (1) FROM EACH SECTION

TIME ALLOTTED: 3 HOURS

SECTION A

QUESTION 1 [17.5 MARKS]

Find the Fourier series of the function, $f(x) = x^2$ on $-\pi < x < \pi$

QUESTION 2 [17.5 MARKS]

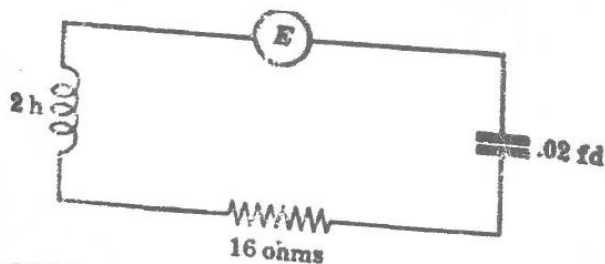
Find the Taylor series for

- a) $f(x) = \sqrt{1+x}$ centered around zero
- b) $f(x) = \sqrt[3]{1+x}$ centered around zero

SECTION B

QUESTION 3 [17.5 MARKS] Using the LaPlace transform method

- a). Solve the equation $\frac{dy}{dt} - 2y = 4$ given that at $t=0$, $y=1$.
- b). an inductor of 2 henrys, a resistor of 16 ohms and a capacitor of .02 farads are connected in series with an e.m.f, of E volts. At $t_0 = 0$ the charge on the capacitor and current in the circuit are zero. Find the charge and current at any time $t > 0$ if $E = 300$ (volts).



QUESTION 4 [17.5 MARKS]

- a). Define the z-transform of a sequence $x(n)$, and find its transform given that $x(n) = a^n u(n)$.
- b). A pipeline system experiences a sudden pressure drop due to a Valve closure. The governing equation for pressure $p(t)$ in the pipeline is given by $\frac{dp}{dt} + ap = b$ where a and b are constants, and the initial pressure before closure is 1 unit. Determine $p(t)$.

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$$(1 - t^2) \frac{d^2 T}{dt^2} - 3t \frac{dT}{dt} + 9T = 0$$

QUESTION 6 [17.5 MARKS]

For the modelling equation relating the depth of an oil well (x) with crude quantity in the well (y) shown below, verify the applicability of the Frobenius method to solve the model. If applicable, find two linearly independent solutions of the modelling equation about the origin.

$$4xy'' + 2y' + y = 0$$

Paper 8 of 8 — GET 311

Level 300 • First Semester • Handout

GET 311 Assignment (due for submission by 3 pm on Tuesday, 05/10/2021)

Use the Frobenius method to find two linearly independent solutions of the following equation about the origin.